LEARNER GUIDE

Inequalities on a number line

Equations, Inequalities and Simultaneous Relationships · Lesson 10. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

An equation like $x = 3$ has one answer. But 'at least 12 to ride' or 'spend at most £10' describes a whole **range** of allowed values — an **inequality**. We write > (more than), < (less than), ≥ (at least) and ≤ (at most), and picture the solution on a **number line**: an **open** circle (■) if the boundary is excluded (> <), a **closed** circle (●) if it is included (≥ ≤), and an arrow for the direction. You solve an inequality with the same steps as an equation — with one new rule: multiplying or dividing by a **negative** reverses the sign.

What you are learning to notice

- An inequality's solution is a **range**, not a single value.
- Solve it like an equation — add, subtract, divide — the same steps.
- The one new rule: × or ÷ by a **negative** flips the sign (> becomes <, and so on).
- Represent it: open circle for > <, closed for ≥ ≤; the relation sets the arrow's direction.

Moving through it, one step at a time

Rearrange the inequality to get x on its own, using the same moves as for an equation. One thing changes: if you multiply or divide both sides by a negative number, every value reflects across 0 (2 and 5 become –2 and –5), which reverses their order — so the inequality sign reverses too. Then draw a number line: put a circle at the boundary — open for a strict inequality (> or <), closed for 'or equal to' (≥ or ≤) — and an arrow in the direction of the allowed values. Read the range back in words to check it makes sense.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — A two-step inequality

Solve $2x + 1 > 7$ and show it on a number line.

1. Subtract 1 from both sides (as for an equation).	$2x > 6$
2. Divide by 2 — positive, so no flip.	$x > 3$
3. Represent it: > means an open circle, pointing right.	open circle at 3, arrow right

$x > 3$ — all values greater than 3 (3 itself excluded, so an open circle).

Example 2 — When the sign flips

Solve $5 - 2x \geq 1$ — watch the flip.

1. Subtract 5 from both sides.	$-2x \geq -4$
2. Divide by –2 — negative, so REVERSE the sign.	$x \leq 2$
3. Represent it: ≤ means a closed circle, pointing left.	closed circle at 2, arrow left

$x \leq 2$ — dividing by –2 flipped ≥ to ≤; 2 is included, so a closed circle.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the number line and directed (negative) numbers, and solving a two-step equation), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Learner Guide v0.1.